ShiftSense — Testing, Acceptance and Pilot Readiness

Version 1.0 · October 2026.

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1. Release status

Level	Status
Demonstration-ready	**Yes**
PoC-verified	**Yes**: H1–H5 pass on synthetic data; full suite passes on SQLite and PostgreSQL 16
Pilot-ready	**Ready for a supervised pilot once the client items in §6 are provided** (hosting, SSO, email relay, real data, validated assumptions, screen-reader session)

2. Automated tests

Run	Result	Files
Full suite, SQLite	**767 passed, 2 skipped** (the 2 PostgreSQL-server tests)	`evidence/test-results/automated-tests/`
Full suite, PostgreSQL 16	**768 passed, 1 skipped** (SQLite-file drill; the PostgreSQL drill ran)	`evidence/test-results/database-compatibility/`

769 test cases in 32 files cover hard rules, Erlang C and forecasting, optimiser and baseline, recovery, workflows end to end, the permission matrix (every action × 6 roles), live/demo isolation, HR decision record, security, photos, imports, PostgreSQL backup/restore/migration, email over SMTP, and Streamlit screen tests for every role. Catalogue: `supporting/testing/workflow-test-catalog.xlsx`. Traceability from requirement to test: `supporting/testing/requirements-traceability.xlsx`.

3. Evidence: hypotheses H1–H5 (simulated)

	Hypothesis	Result	Measured
H1	Planning effort falls ≥ 70 %	Pass	92 % less (400 → 30 min, modelled); coverage 90.9 % vs 75.4 %
H2	Under-covered workload falls ≥ 20 % at equal or lower paid hours	Pass	24.6 % → 9.1 %; paid hours 3,824 vs 3,824

	Hypothesis	Result	Measured
H3	Median Time-to-Cover < 10 min	Pass	7.0 min vs 58 min manual (modelled)
H4	Equal coverage at lower cost than overtime-first	Pass	₹4,950 vs ₹9,500 median; cheaper in 8/8
H5	Zero hard-rule violations	Pass	0 across 3 plans, 1 re-plan, 32 recoveries

Ablation (simulated):

Approach	Coverage %	Under-covered %	Fairness std	Re-plan changes	Time-to-Cover (min)	Recovery cost ₹	Violations
Manual heuristic	75.4	24.6	1.85	2	48	4,500	0
Optimiser only	90.9	9.1	2.22	475	60	9,375	0
+ fairness + stability	90.9	9.1	1.60	23	58	9,500	0
ShiftSense full loop	90.9	9.1	1.60	23	7.0	8,800	0

The manual heuristic's lower recovery cost restores only 54 % of the gap (median); the full loop restores 96 %. At equal restored coverage, ShiftSense is cheaper (H4). CSVs: `evidence/test-results/workflow-validation/`.

4. Accessibility

Automated checks in headless Chrome on 31 screens of all six roles at 1440, 768 and 375 px (93 checks): accessible names for every button, link and input; images with alt text; page language set; one H1 per screen; no sideways page scrolling. Results: `evidence/test-results/accessibility/`. Contrast is tested in `test_app.py` (muted text ≥ 4.5:1, body text ≥ 7:1, both themes). **Not done:** a session with a real screen reader (NVDA / VoiceOver).

5. Workflow validation

- End-to-end workflows for every role are automated in `test_scenarios_final.py`, `test_workflows_e2e.py`, `test_acceptance.py` and the HR suites (see `evidence/test-results/workflow-validation/README.md`).
- In a real browser, on a throwaway copy of the database, these were exercised: publish, rollback, swap (agent → colleague → team lead), HR separation with second-user confirmation, HR policy, governance scan, theme switching, and the coverage matrix and calendars at desktop and phone widths.
- Screenshots for every role in light and dark themes: `evidence/screenshots/`.

Known issues and limitations

Status as of 21 Oct 2026. Release status: **demonstration-ready and PoC-verified; pilot needs the items in §2.**

1. By design (PoC scope)

- All data is synthetic (seeded). Costs use configured simulated rates; recovery outcomes are simulated.
- No trained ML: the forecast is a seasonal average; plans come from the CP-SAT optimiser.
- Forecast accuracy and actual service level show "Unavailable" until real call volumes are imported.

- No real integrations (payroll, WFM, telephony); data comes in by file or form.
- One application server. Workspace saves are versioned (a conflicting save is refused), but multi-server deployment has not been tested.
- Retention periods are not set (HR-08 deferred); nothing is deleted automatically.
- Absence flags and overtime approval are built but **off** until thresholds are approved.

2. Needs the client / a person before a pilot

Item	What is needed
Single sign-on	Identity-provider client ID and secret; built and unit-tested, not run against a real provider
Email delivery	A real SMTP relay; tested only against a local SMTP server. "Sent" = accepted by relay, not delivered
Hosting	A host for the Docker stack (PostgreSQL 16, app, jobs, Caddy). Compose file validated; full stack not run on a server
Screen-reader session	NVDA / VoiceOver session by a person; automated accessibility checks pass
Demo accounts	Change passwords or remove accounts (health check fails in pilot mode otherwise)
Assumptions	Validate costs, recovery timings, governance thresholds, HR thresholds with EXL
Restore rehearsal	Restore on a production-like environment (automated drill passes on PostgreSQL)
Real HRIS export	Leave-balance feed handles flexible headers; the real file format has not been seen

3. Known technical limits

- **Restart after code changes**: Streamlit runs with `fileWatcherType = "none"`.
- **Theme switch** relies on Streamlit 1.64 menu identifiers (`stMainMenuItem-theme-*`); pinned in `requirements.txt`, re-check on upgrade.
- No icon-only sidebar rail (Streamlit's own collapse control is used).
- The coverage matrix is dense on very small phones at 1-hour interval; the 2-hour interval shows full cell text.
- Roster calendars are view-only (no drag-and-drop); roster changes are explicit, permission-checked actions.
- Screens do not auto-refresh; use **Refresh** in the sidebar.
- Larger sites than the synthetic set (50 people, 4 queues; full pack 120 people, 6 queues) were not load-tested.

Pilot acceptance checklist

Status words: **Done**: verified by automated tests and/or in a browser. **Client**: needs something only the client can provide. **Open**: not done yet.

A. Functional (PoC scope)

#	Criterion	Evidence	Status
A1	Roster built with zero hard-rule violations	H5; `tests/test_constraints.py`; health check	Done
A2	Optimised plan beats the manual heuristic on coverage at equal paid hours	H1, H2 (simulated)	Done
A3	Absence → ranked fix → team-lead decision → roster updated, residual risk escalated	H3, H4; `test_workflows_e2e.py`	Done

#	Criterion	Evidence	Status
A4	Leave Green/Amber/Red with alternatives; HR review triggers	`test_workflow.py`, `test_hr_record.py`	Done
A5	Every action permission-checked on the server (6 roles)	`test_matrix.py`; role-permission-matrix.xlsx	Done
A6	Demo scenario never changes live data	`test_isolation.py`	Done
A7	Audit trail append-only and hash-chained	`test_services.py`; health check	Done
A8	HR decision record HR-01..HR-08	`test_hr_record.py`, `test_hr_gaps.py`	Done
A9	Imports validate and never invent mandatory fields	`test_data_system.py`, `test_importer.py`	Done
A10	Light and dark themes, role navigation, calendars, coverage matrix	`test_ui_overhaul.py`; screenshots	Done

B. Quality

#	Criterion	Evidence	Status
B1	Full suite passes on SQLite	767 passed, 2 skipped (PostgreSQL-only)	Done
B2	Full suite passes on PostgreSQL 16	768 passed, 1 skipped (SQLite-only drill)	Done
B3	Automated accessibility checks (names, labels, alt, lang, headings, no sideways scroll at 1440/768/375 px)	`evidence/test-results/accessibility`	Done
B4	Screen-reader session (NVDA / VoiceOver)	needs a person	Open
B5	Manual browser runs: publish, rollback, swap, separation, HR policy, governance	READINESS pass 19 Oct 2026 (throwaway copy)	Done

C. Before go-live (client)

#	Item	Status
C1	Host for the Docker stack; DNS name for HTTPS (`SHIFTSENSE_DOMAIN`)	Client
C2	Identity provider (OIDC) client ID and secret; `SHIFTSENSE_AUTH=sso` or `both`	Client
C3	SMTP relay host and credentials; send the test email and check the inbox	Client
C4	Remove demo accounts or change their passwords; create real accounts	Client
C5	Validate assumptions: pay rates, shrinkage, SLA targets, recovery timings, governance and HR thresholds	Client
C6	Approve retention periods (HR-08) and absence-flag / overtime thresholds	Client
C7	Real data: employee export, call-volume history, HRIS leave-balance export	Client
C8	Restore rehearsal on the pilot host (`scripts/backup.sh drill` , then a real restore test)	Client
C9	Security review / penetration test	Client
C10	`scripts/health.sh` passes in pilot mode (`SHIFTSENSE_ENV=pilot`)	Open until C1–C4

Sign-off

Role	Name	Date	Signature
Business owner (EXL)			

Role	Name	Date	Signature
Operations lead			
HR			
IT / security			